

RWD Lineage

A traceability standard to trust SDTM generated from RWD for regulatory decision-making



2026-03-30

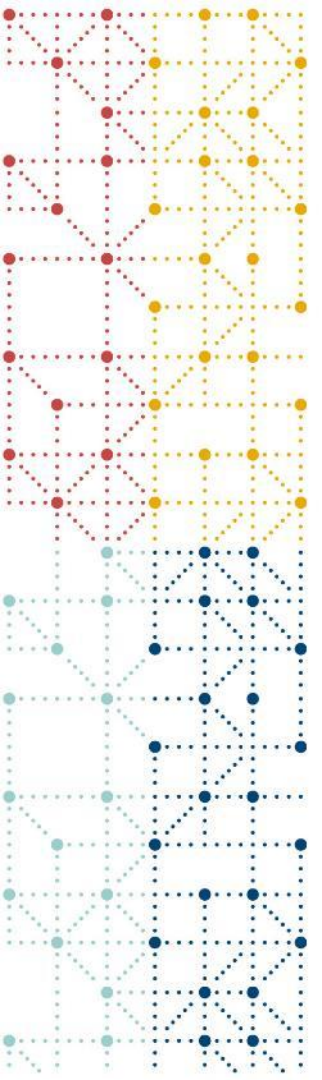
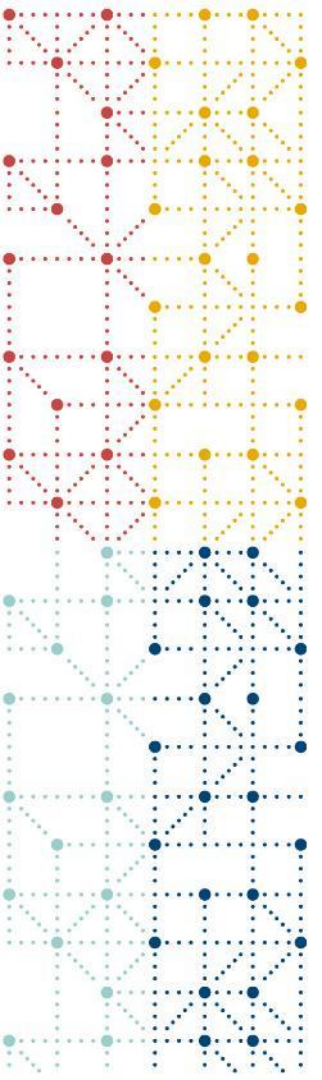


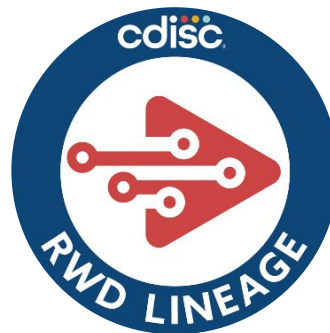
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RWD Lineage: Background & Motivation

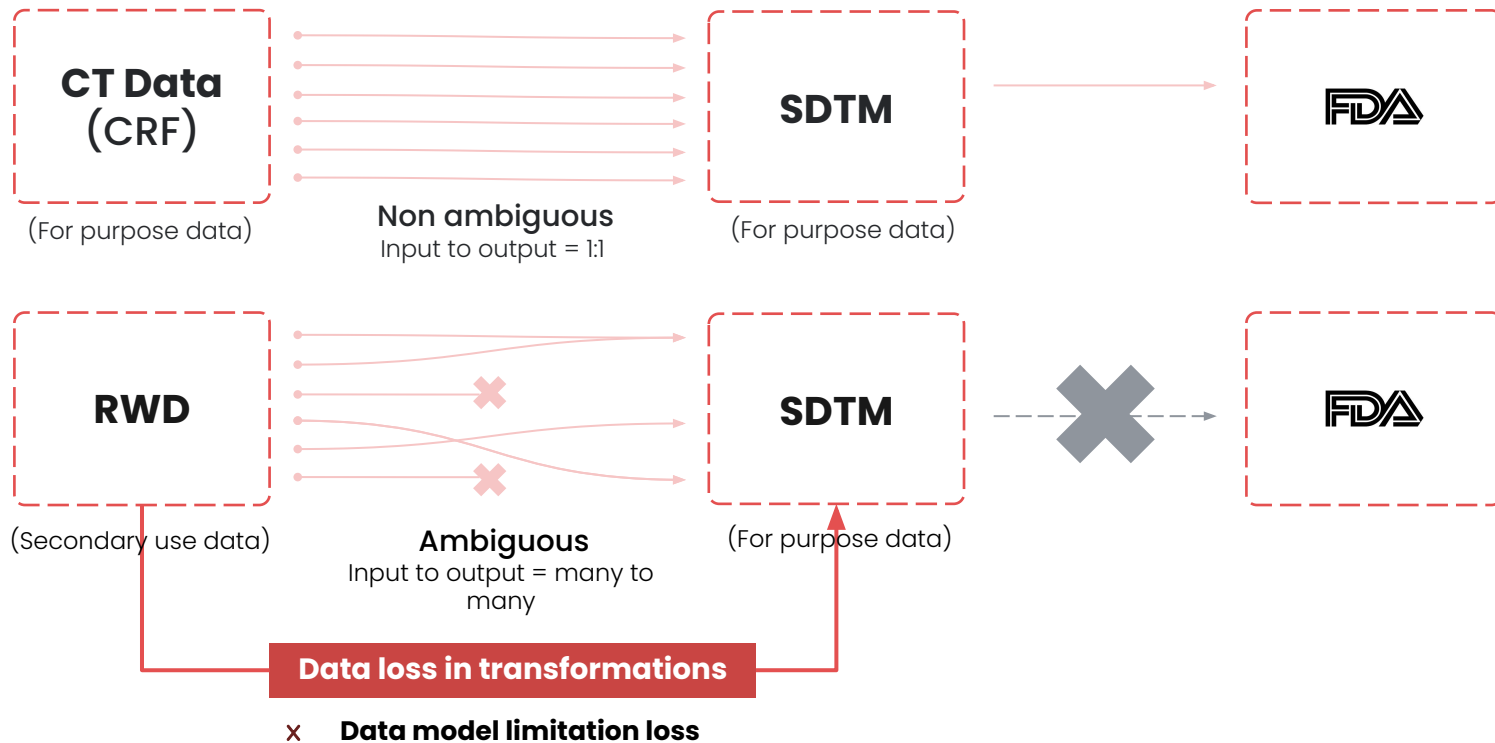
Introducing RWD Lineage



Project Goal

Create a CDISC data exchange standard for lineage metadata that is supplied along with RWD-derived SDTM, which provides the data reliability required by FDA to use RWE as primary evidence.

Reliability in RWD: Challenges



FDA: No Lowered Quality Standards for RWD/RWE



Real-World Data: Assessing
Electronic Health Records and
Medical Claims Data to Support
Regulatory Decision-Making
for Drug and Biological
Products
Guidance for Industry

July 2024

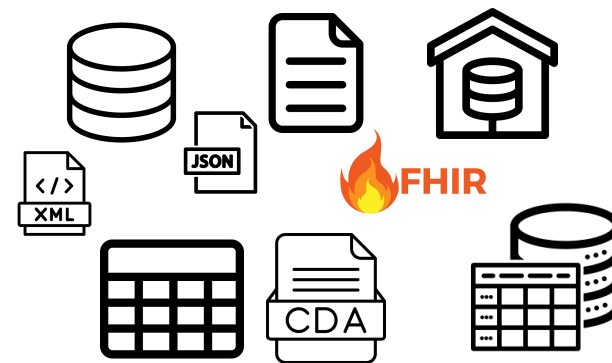
“For all study designs, it is important to ensure the reliability... of the data used to help support a regulatory decision. For the purposes of this guidance, the term reliability includes accuracy, completeness, and traceability.”

RWD: Challenges for FDA Review

When reviewing and inspecting RWD submission, FDA needs to ***interrogate source data*** to determine ***reliability*** of study data



Validation & verification



Disparate RWD data sources

Reliability: Atomic Lineage for Individual Data Points

SDTM

MH						
...	USUBJID	MHSEQ	MHTERM	MHPRESP	MHOCCUR	...
...	001	1	HISTORY OF MYOCARDIAL INFARCTION	Y	Y	...
...	001	2	TYPE 2 DIABETES	Y	N	...
...	001	3	HYPERTENSION	Y	Y	...
...	...	...	...	...	...	...
...	002	15	HYPERTENSION	Y	Y	...
...	...	...	...	...	...	...
...	003	27	HYPERTENSION	Y	N	...
...	...	...	...	...	...	...



Real-World Data: Assessing
Electronic Health Records and
Medical Claims Data to Support
Regulatory Decision-Making
for Drug and Biological
Products
Guidance for Industry

July 2024

“...sponsors should evaluate the completeness, accuracy, and plausibility of the data, including verifying data against its original source (e.g., discharge notes, pathology reports, registry records)...”

“...subject matter experts’ review of medical records (including structured and unstructured data) may be a preferred reference standard for validation of clinical events identified by diagnosis codes or automated algorithms...”

Reliability: Atomic Lineage for Individual Data Points

SDTM

MH						
...	USUBJID	MHSEQ	MHTERM	MHPRESP	MHOCCUR	...
...	001	1	HISTORY OF MYOCARDIAL INFARCTION	Y	Y	...
...	001	2	TYPE 2 DIABETES	Y	N	...
...	001	3	HYPERTENSION	Y	Y	...
...	...	...	...	...	...	...
...	002	15	HYPERTENSION	Y	Y	...
...	...	...	...	...	...	...
...	003	27	HYPERTENSION	Y	N	...
...	...	...	...	...	...	...

*Source
RWD*

PT_DX		
PT_ID	ICD10	TERM
21962	I10	Essential hypertension
21962	N18.3	Chronic kidney disease, stage 3
...	...	...

Reliability: Atomic Lineage for Individual Data Points

SDTM

MH					
...	USUBJID	MHSEQ	MHTERM	MHPRESP	MHOCCUR
...	001	1	HISTORY OF MYOCARDIAL INFARCTION	Y	Y
...	001	2	TYPE 2 DIABETES	Y	N
...	001	3	HYPERTENSION	Y	Y
...	...	...	...	...	...
...	002	15	HYPERTENSION	Y	Y
...	...	...	...	...	...
...	003	27	HYPERTENSION	Y	N
...	...	...	...	...	...

*Source
RWD*

PT_DX		
PT_ID	ICD10	TERM
21962	I10	Essential hypertension
21962	N18.3	Chronic kidney disease, stage 3
...	...	...

VITALS		
Patno	Vital	Value
19251	BP	150/110
19251	BMI	28.3
...	...	...

NOTES	
PT_ID	TEXT
19251	Medical history: Type 2 DM on insulin, CHF, HTN CKD3, ...
19251	Discharge summary: ...
...	...

Reliability: Atomic Lineage for Individual Data Points

SDTM

MH						
...	USUBJID	MHSEQ	MHTERM	MHPRESP	MHOCCUR	...
...	001	1	HISTORY OF MYOCARDIAL INFARCTION	Y	Y	...
...	001	2	TYPE 2 DIABETES	Y	N	...
...	001	3	HYPERTENSION	Y	Y	...
...	...	...	...	...	...	...
...	002	15	HYPERTENSION	Y	Y	...
...	...	...	...	...	...	...
...	003	27	HYPERTENSION	Y	N	...
...	...	...	...	...	...	...

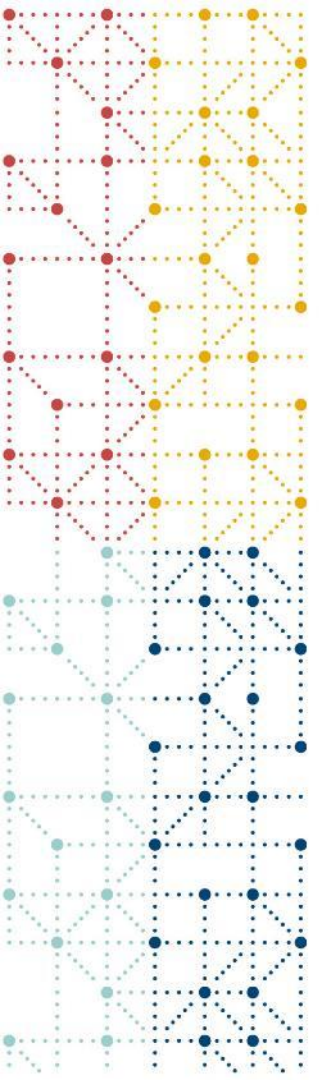
True positive

True positive

False negative

Interrogate source data to validate the accuracy of individual data points

Atomic lineage provides traceability of individual data points, including coordinates and values, from target SDTM to original source data



RWD Lineage in Define-XML

Solution: RWD Lineage in Define-XML

- Define-XML: already in the data standards catalog
- Minimal impact to existing workflows
 - Self-contained lineage metadata references in Define-XML
 - Lineage metadata can be stored in standalone files
- Easy and pain-free adoption



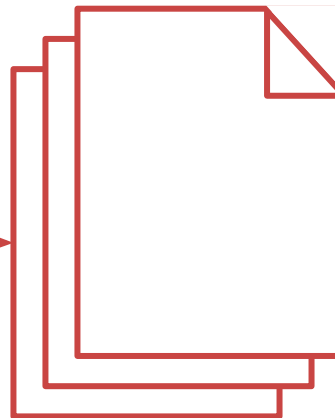
Solution: RWD Lineage in Define-XML

Self-contained `<rwdl:lineage>`
XML tag referencing RWD
Lineage .XML files

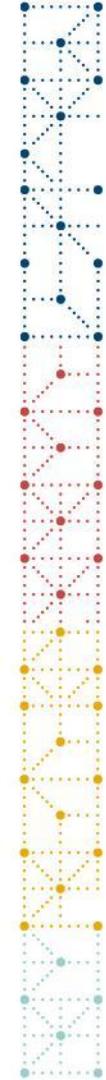
```
<define>  
  
  <rwdl:Lineage>  
    ...  
  </rwdl:Lineage>  
  
</define>
```

Define-XML

Detailed data point level lineage
metadata stored in one or more XML
files referenced in Define-XML



RWD Lineage



RWD Lineage to support primary evidence

e.g., external controls, pragmatic trials

Standard data point-level lineage enables FDA to easily:

- Inspect the individual data points in source data for any data point in SDTM
 - Verify & validate any data point against the source
 - Answer questions like: Is the patient truly exposed to the study treatment? Do terminology mappings match source values? etc.
- Validate accuracy of computable phenotypes against source data
- Ensure that SDTM is fully traceable to source data

Result:

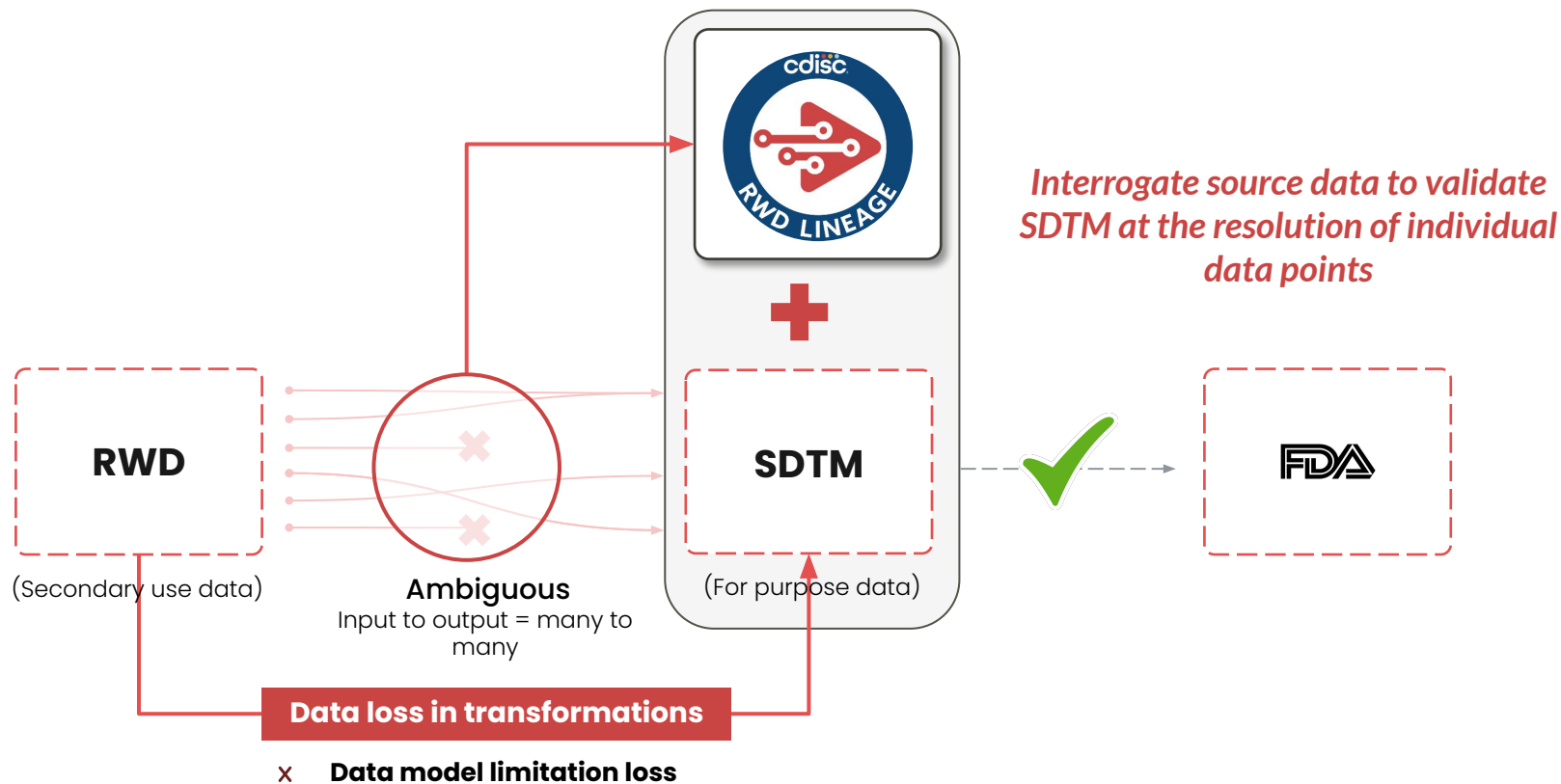
- **Ability to easily review & audit SDTM derived from RWD**
- **Trust evidence derived from RWD**

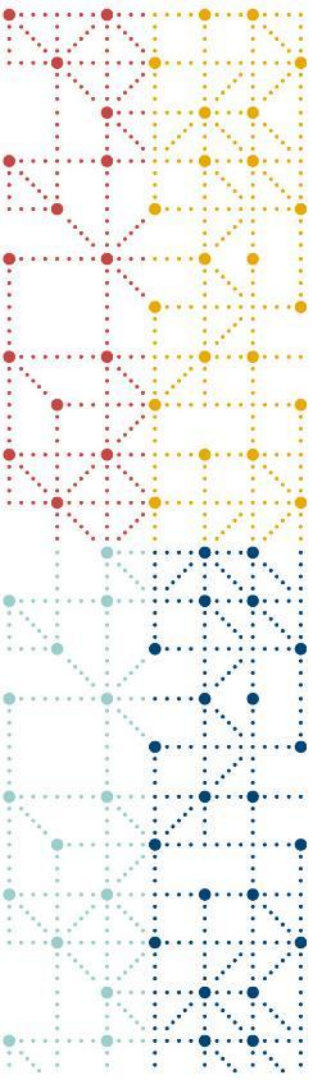
Key requirements

- Data-point level lineage
- Can “stack” multiple stages of transformations
 - Example: FHIR > CDM > SDTM
 - Example: Multiple aggregated EHR sources > SDTM
- Supports commonly used formats of RWD
 - Tabular data: e.g., EHR, claims, aggregated EHR in common data models
 - Tree-based formats: e.g., FHIR (JSON), CCDa (XML)
 - Files: e.g., documents, PDF charts, images/signals (file-level)

Takeaway: RWD Lineage standard will cover all the general use cases of RWD in regulatory decision-making

RWD LINEAGE for Reliable RWE





Implementation Examples

RWD-Lineage Specification Overview

RWD Lineage

A machine-readable XML extension to Define-XML that captures **data point-level lineage** metadata — tracing every value in an RWD-derived SDTM dataset back to its exact source in the originating EHR, lab system, or clinical data store.

Why?

FDA and other regulators need data reliability evidence to accept RWE as primary evidence. RWD-Lineage provides that audit trail in a standardized, machine-readable format.

Implemented as a namespace extension to Define-XML. The `rwdl:lineage` root wraps `rwdl:sourceMetadata` (source assertions) and `rwdl:lineageTrail` (the `MapID` array) as independent siblings.

`<rwdl:Lineage>`

Root document — two children

`<rwdl:
LineageTrail>`

The forensic record —
the array of `<MapID>`
mappings that trace
every value.

`<rwdl:
SourceMetadata>
• optional`

Annotation layer —
independent of the trail.

Inside the Lineage Trail

Implementation in Define-XML

Implemented as a namespace extension to Define-XML (rwdl:lineage), provided inside Define-XML or referenced as a standalone XML file alongside SDTM datasets.

- Source - source data point (e.g., in EHR/FHIR/...)
- Target - target data point (e.g., in SDTM)
- UUID - unique identifier for a source-target data point pair
- MethodDefOID - a reference to Define-XML MethodDef to describe the transformation between the source-target pair

Lineage Trail Unit: <MapID>

UUID	Deterministic UUIDv5 from Source + Target hash
MethodDefOID	OID ref to a Define-XML MethodDef (transform logic)
rwdl:Source	Coordinate of the origin RWD data point
rwdl:Target	Coordinate of the destination SDTM cell

Coordinate = Storage x Structure

Database | Filesystem | API | Message x Tabular | Path | Object

Draft specification in GitHub:

https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/documents/RWD-Lineage_Data_Standard_Specification.md

Worked Examples

Data-point level lineage across different clinical data transformation scenarios

Example 1

Clinical Events (CE)

EHR diagnoses, vitals & clinical notes → SDTM CE domain

Traceability scenario:

Multi-source evidence convergence

A single SDTM cell (CEOCCUR) traces back to source data points with ICD-10 diagnosis codes, blood pressure measurements, and NLP-extracted findings from clinical notes — all contributing to one clinical event determination.

3 source tables • 2 subjects • CE domain

Example 2

Labs & Adverse Events (LB, AE)

EHR LOINC lab results → SDTM LB + AE domains

Traceability scenario:

Cross-domain derivation chain

Raw lab result strings are parsed, converted to standard units, and evaluated against reference ranges in LB. Elevated values then derive an adverse event record in AE — full traceability from data points in source EHR through both domains.

1 source table • 2 subjects • LB + AE domains

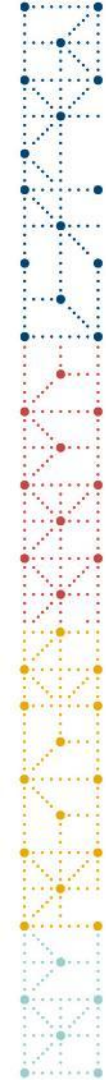
Example 3

Coming Soon

Additional scenarios under development to demonstrate lineage across other RWD source formats and SDTM domains.

Potential scenarios:

- Multiple data sources
- Larger scale example using real EHR data
- FHIR → SDTM (tree-structured source)
- ...



Key Concepts Demonstrated

Data Point-Level Traceability

Every individual data point in SDTM output maps to a specific row and column in the source data — not table-level or column-level, but data point-level granularity.

Multi-Source Evidence

A single SDTM value (e.g., CEOCCUR) can trace to multiple source tables simultaneously — diagnoses, vitals, and NLP findings converge on one determination.

Transformation Provenance

Each mapping records the transformation type applied (DirectMap, UnitConversion, NLPExtraction, etc.), documenting the logic chain from source to target.

Lineage Stackability Across Multiple Transformations

Derived domains (e.g., AE from LB) are fully traceable — the lineage captures both the derivation step and the original source-to-intermediate mapping.

Takeaway: RWD-Lineage closes the traceability gap between raw real-world data and regulatory-grade SDTM — enabling FDA to trust RWE as primary evidence.

Important Links

RWD Lineage GitHub:

<https://github.com/cdisc-org/DataExchange-RWD-Lineage/tree/main>

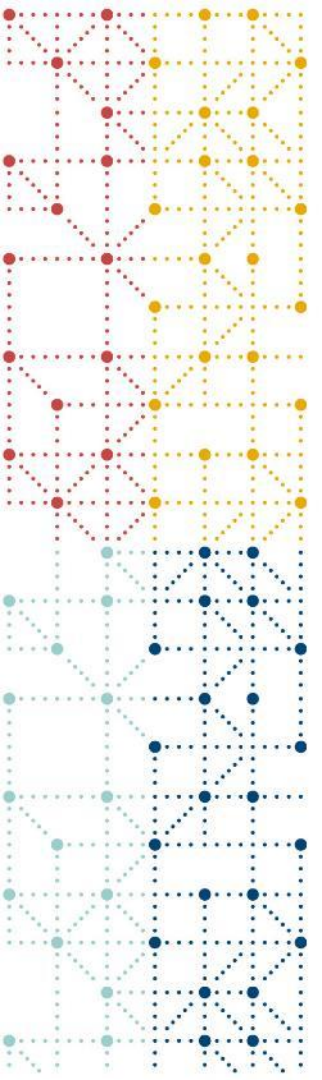
Working examples of RWD Lineage specification and implementation examples

RWD Lineage wiki: <https://wiki.cdisc.org/display/RWDLIN/RWD+Lineage>

Project wiki

Disclaimer: The RWD Lineage standard is WIP. Feedback or any issue reports are welcome and appreciated in the GitHub/wiki, or as email to the project coleads

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Tasha Nagamine: tasha@droicelabs.com



Example 1: Hypertension and Myocardial Infarction Clinical Events (CE)

Example 1: Hypertension and Myocardial Infarction Clinical Events

EHR diagnoses, vitals & clinical notes → SDTM CE domain

Scenario

Two prespecified clinical events — hypertension and acute myocardial infarction — are tracked across two subjects. Each event's occurrence (CEOCCUR) is determined by a multi-source algorithm drawing on ICD-10 diagnosis codes, blood pressure readings, and NLP-extracted findings from clinical notes.

Algorithms

Hypertension

- ICD-10 code I10–I16 (Hypertensive diseases)
- ≥ 2 elevated BP measurements over 3 months
- Diagnosed hypertension in clinical notes

Acute Myocardial Infarction

- Diagnosis code I21 or I22 (STEMI / NSTEMI MI)
- Records of MI in clinical notes within follow-up

Source EHR Tables

Source Table	Description
pt_dx.csv	ICD-10 diagnosis codes
vitals.csv	Blood pressure, BMI
notes.csv	Free-text clinical notes

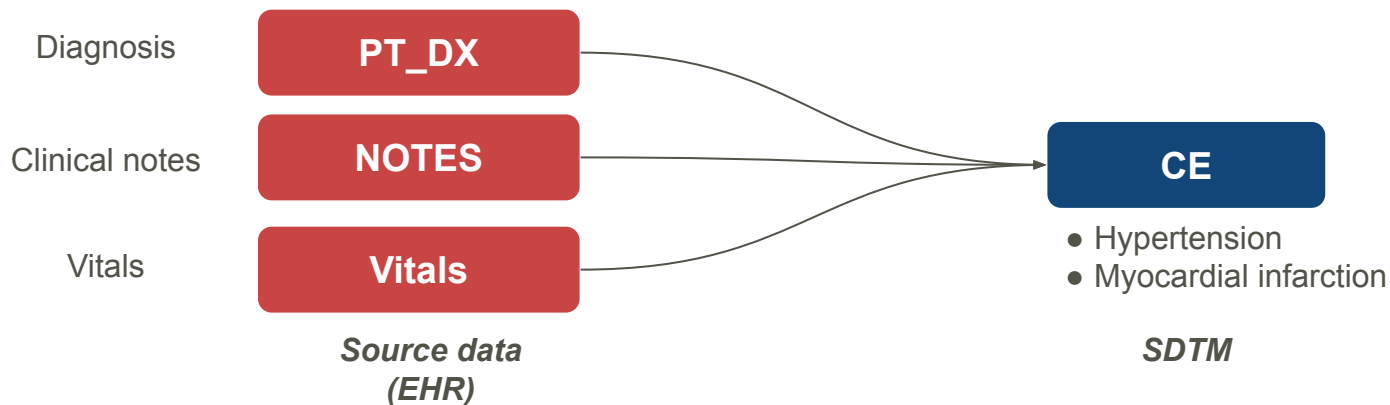
Target: ce.csv (SDTM CE — 4 records)

SDTM Output (CE)

USUBJID	CETERM	CEOCCUR
001	MYOCARDIAL INFARCTION	Y
001	HYPERTENSION	Y
002	MYOCARDIAL INFARCTION	N
002	HYPERTENSION	Y

All lineage entries target CEOCCUR — the occurrence flag is the outcome of multi-source evidence evaluation

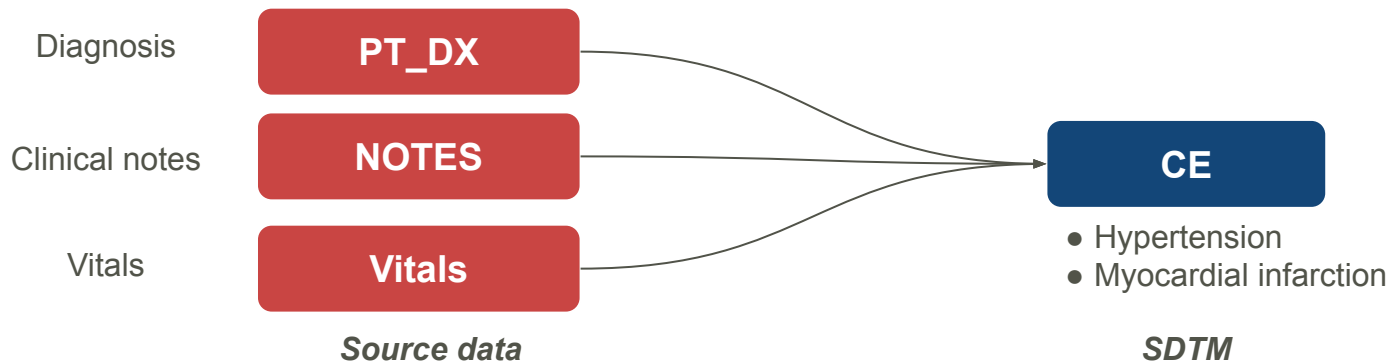
Example 1: SDTM CE Derivation



Two prespecified clinical events:

- Hypertension
 - Patient is assigned ICD10 diagnosis code I10–I16 (Hypertensive diseases)
 - Patient has at least two elevated blood pressure measurements ($\geq 130/80$ mmHg) over the course of a 3-month period within the followup period
 - Patient has diagnosed hypertension in clinical notes
- Myocardial infarction
 - Patient is assigned ICD10 diagnosis code I21 (Acute myocardial infarction) or I122 (Subsequent ST elevation (STEMI) and non-ST elevation (NSTEMI) myocardial infarction)
 - Patient has records of acute myocardial infarction in notes within the followup period

Example 1: SDTM CE Derivation



Transformation from source to SDTM

PT_DX, NOTES, Vitals → CE

RWD Lineage from SDTM to source

CE → PT_DX, NOTES, Vitals

Example 1: Subject 001 Hypertension

SDTM

CE				
USUBJID	CESEQ	CETERM	CEPRES	CEOCCUR
001	1	MYOCARDIAL INFARCTION	Y	Y
001	2	HYPERTENSION	Y	Y
002	1	MYOCARDIAL INFARCTION	Y	N
002	2	HYPERTENSION	Y	Y

*Source
RWD*

PT_DX			
PT_ID	DATE	ICD10	TERM
21962	05/30/2025	I21.3	ST elevation (STEMI) myocardial infarction of unspecified site
21962	06/27/2025	N18.3	Chronic kidney disease, stage 3
21962	06/27/2025	I25.2	Old myocardial infarction
21962	06/27/2025	I10	Essential hypertension
19251	04/16/2024	E11.9	Type 2 diabetes mellitus without complications
19251	05/28/2024	E11.65	Type 2 diabetes mellitus with hyperglycemia
19251	07/25/2024	E11.65	Type 2 diabetes mellitus with hyperglycemia
19251	07/25/2024	K80.11	Calculus of gallbladder with chronic cholecystitis with obstruction

Example 1: Subject 001 Hypertension

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <rwdl:Lineage xmlns:rwdl="http://www.cdisc.org/ns/rwdl/v1.0">
3
4     <!-- LAYER 1: Source metadata - assertions about the source systems -->
5     <rwdl:SourceMetadata>
6         <rwdl:SourceSystem OID="SRC.CSV.1"
7             Name="Example 1 Clinical Source CSV Files"
8             Description="CSV exports from clinical source system; includes patient diagnoses (pt_dx.csv), vitals (vitals.csv), and clinical notes
9             <rwdl:ExternalCodeList Dictionary="ICD-10-CM" Version="2024"
10                 AppliesTo="pt_dx.csv ICD10"/>
11         </rwdl:SourceSystem>
12     </rwdl:SourceMetadata>
13
14     <!-- LAYER 2: Lineage trail - Source -> Target pairs -->
15     <rwdl:LineageTrail>
16
17         <!-- pt_dx.csv row 4 -> ce.csv row 2 -->
18         <rwdl:MapID UUID="35060134-fc2f-4cdf-9abe-491924739bd5">
19             <rwdl:Source>
20                 <rwdl:Coordinate Storage="FILESYSTEM" Structure="TABULAR" Format="CSV">
21                     <rwdl:URI>https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/source/pt_dx.csv</rwdl:URI>
22                     <rwdl:RowIndex>4</rwdl:RowIndex>
23                     <rwdl:ColumnName>ICD10</rwdl:ColumnName>
24                 </rwdl:Coordinate>
25             </rwdl:Source>
26             <rwdl:Target>
27                 <rwdl:Coordinate Storage="FILESYSTEM" Structure="TABULAR" Format="CSV">
28                     <rwdl:URI>https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/tdm/ce.csv</rwdl:URI>
29                     <rwdl:RowIndex>2</rwdl:RowIndex>
30                     <rwdl:ColumnName>CEOCCUR</rwdl:ColumnName>
31                 </rwdl:Coordinate>
32             </rwdl:Target>
33         </rwdl:MapID>
```

← 4th row of column ICD10 in table PT_DX

maps to

← 2nd row of column CEOCCUR in table CE

Example 1: Subject 001 Hypertension

```
31         </rwdl:Coordinate>
32     </rwdl:Target>
33 </rwdl:MapID>
34
35 <rwdl:MapID UUID="77d5b278-692e-4d9b-8232-179b34f8e6b3">
36     <rwdl:Source>
37         <rwdl:Coordinate Storage="FILESYSTEM" Structure="TABULAR" Format="CSV">
38             <rwdl:URI>https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/source/pt_dx.csv</rwdl:URI>
39             <rwdl:RowIndex>4</rwdl:RowIndex>
40             <rwdl:ColumnName>TERM</rwdl:ColumnName>
41         </rwdl:Coordinate>
42     </rwdl:Source>
43     <rwdl:Target>
44         <rwdl:Coordinate Storage="FILESYSTEM" Structure="TABULAR" Format="CSV">
45             <rwdl:URI>https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/tdm/ce.csv</rwdl:URI>
46             <rwdl:RowIndex>2</rwdl:RowIndex>
47             <rwdl:ColumnName>CEOCCUR</rwdl:ColumnName>
48         </rwdl:Coordinate>
49     </rwdl:Target>
50 </rwdl:MapID>
51
52 <rwdl:MapID UUID="7e376beb-7dad-4f5c-a212-88283ac22eba" MethodDefOID="MT.AFTERIDXDATE">
53     <rwdl:Source>
54         <rwdl:Coordinate Storage="FILESYSTEM" Structure="TABULAR" Format="CSV">
55             <rwdl:URI>https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/source/pt_dx.csv</rwdl:URI>
```

← 4th row of column TERM in table PT_DX

maps to

← 2nd row of column CEOCCUR in table CE

Example 1: Subject 001 Hypertension

```
47         <rwdl:ColumnName>CEOCCUR</rwdl:ColumnName>
48     </rwdl:Coordinate>
49 </rwdl:Target>
50 </rwdl:MapID>
51
52 <rwdl:MapID UUID="7e376beb-7dad-4f5c-a212-88283ac22eba" MethodDefOID="MT.AFTERIDXDATE"> ← Comparison to index date
53     <rwdl:Source>
54         <rwdl:Coordinate Storage="FILESYSTEM" Structure="TABULAR" Format="CSV">
55             <rwdl:URI>https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/source/pt_dx.csv</rwdl:URI>
56             <rwdl:RowIndex>4</rwdl:RowIndex>
57             <rwdl:ColumnName>DATE</rwdl:ColumnName> ← 4th row of column DATE in table PT_DX
58         </rwdl:Coordinate>
59     </rwdl:Source>
60     <rwdl:Target>
61         <rwdl:Coordinate Storage="FILESYSTEM" Structure="TABULAR" Format="CSV">
62             <rwdl:URI>https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/sdtm/ce.csv</rwdl:URI>
63             <rwdl:RowIndex>2</rwdl:RowIndex>
64             <rwdl:ColumnName>CEOCCUR</rwdl:ColumnName> ← 2nd row of column CEOCCUR in table CE
65         </rwdl:Coordinate>
66     </rwdl:Target>
67 </rwdl:MapID>
68
69 <!-- vitals.csv row 3 -> ce.csv row 4 -->
70 <rwdl:MapID UUID="880af520-a197-44c7-976c-3770bdc46575">
71     <rwdl:Source>
72         <rwdl:Coordinate Storage="FILESYSTEM" Structure="TABULAR" Format="CSV">
73             <rwdl:URI>https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/source/vitals.csv</rwdl:URI>
```

maps to

Example 1: Subject 002 Myocardial infarction

SDTM

CE				
USUBJID	CESEQ	CETERM	CEPRES	CEOCCUR
001	1	MYOCARDIAL INFARCTION	Y	Y
001	2	HYPERTENSION	Y	Y
002	1	MYOCARDIAL INFARCTION	Y	N
002	2	HYPERTENSION	Y	Y

**Source
RWD**

PT_DX			
PT_ID	DATE	ICD10	TERM
21962	05/30/2025	I21.3	ST elevation (STEMI) myocardial infarction of unspecified site
21962	06/27/2025	N18.3	Chronic kidney disease, stage 3
21962	06/27/2025	I25.2	Old myocardial infarction
21962	06/27/2025	I10	Essential hypertension
19251	04/16/2024	E11.9	Type 2 diabetes mellitus without complications
19251	05/28/2024	E11.65	Type 2 diabetes mellitus with hyperglycemia
19251	07/25/2024	E11.65	Type 2 diabetes mellitus with hyperglycemia
19251	07/25/2024	K80.11	Calculus of gallbladder with chronic cholecystitis with obstruction

Example 1: Subject 001 Hypertension

```
277 <!-- pt_dx.csv row 1 --> ce.csv row 1 -->
278 <rwidl:MapID UUID="0f185720-823f-4ee3-9e74-2cade2791ed5">
279   <rwidl:Source>
280     <rwidl:Coordinate Storage="FILESYSTEM" Structure="TABULAR" Format="CSV">
281       <rwidl:URI>https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/source/pt_dx.csv</rwidl:URI>
282       <rwidl:RowIndex>1</rwidl:RowIndex>
283       <rwidl:ColumnName>ICD10</rwidl:ColumnName>
284     </rwidl:Coordinate>
285   </rwidl:Source>
286   <rwidl:Target>
287     <rwidl:Coordinate Storage="FILESYSTEM" Structure="TABULAR" Format="CSV">
288       <rwidl:URI>https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/rdm/ce.csv</rwidl:URI>
289       <rwidl:RowIndex>1</rwidl:RowIndex>
290       <rwidl:ColumnName>CEOCCUR</rwidl:ColumnName>
291     </rwidl:Coordinate>
292   </rwidl:Target>
293 </rwidl:MapID>
```

← 1st row of column ICD10 in table PT_DX

maps to

← 1st row of column CEOCCUR in table CE

etc.

```
295 <rwidl:MapID UUID="de098b66-7ac9-4695-ab1a-bf4beea45a8d">
296   <rwidl:Source>
297     <rwidl:Coordinate Storage="FILESYSTEM" Structure="TABULAR" Format="CSV">
298       <rwidl:URI>https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/source/pt_dx.csv</rwidl:URI>
299       <rwidl:RowIndex>1</rwidl:RowIndex>
300       <rwidl:ColumnName>TERM</rwidl:ColumnName>
301     </rwidl:Coordinate>
302   </rwidl:Source>
```

Example 1: Subject 002 Myocardial infarction

SDTM

CE				
USUBJID	CESEQ	CETERM	CEPRES	CEOCCUR
001	1	MYOCARDIAL INFARCTION	Y	Y
001	2	HYPERTENSION	Y	Y
002	1	MYOCARDIAL INFARCTION	Y	N
002	2	HYPERTENSION	Y	Y

*Source
RWD*

NOTES		
PT_ID	DATE	TEXT
21962	06/27/2025	PAST MEDICAL HISTORY: 1. Stage III CKD with baseline creatinine in the 2 range. 2. Status post MI on May 30, 2024.
19251	04/16/2024	PAST MEDICAL HISTORY: Type 2 DM, hypothyroidism, hyperlipidemia, HTN, obstructive sleep apnea
19251	07/25/2024	ADMISSION DIAGNOSES: Hyperglycemia, cholelithiasis, obstructive sleep apnea, diabetes mellitus, and hypertension. DISCHARGE DIAGNOSES: Hyperglycemia, cholelithiasis, obstructive sleep apnea, diabetes mellitus, hypertension, and cholecystitis.

Example 1: Subject 003 Hypertension

SDTM

CE				
USUBJID	CESEQ	CETERM	CEPRES	CEOCCUR
001	1	MYOCARDIAL INFARCTION	Y	Y
001	2	HYPERTENSION	Y	Y
002	1	MYOCARDIAL INFARCTION	Y	N
002	2	HYPERTENSION	Y	Y

**Source
RWD**

NOTES		
PT_ID	DATE	TEXT
21962	06/27/2025	PAST MEDICAL HISTORY: 1. Stage III CKD with baseline creatinine in the 2 range. 2. Status post MI on May 30, 2024 .
19251	04/16/2024	PAST MEDICAL HISTORY: Type 2 DM, hypothyroidism, hyperlipidemia, HTN, obstructive sleep apnea
19251	07/25/2024	ADMISSION DIAGNOSES: Hyperglycemia, cholelithiasis, obstructive sleep apnea, diabetes mellitus, and hypertension. DISCHARGE DIAGNOSES: Hyperglycemia, cholelithiasis, obstructive sleep apnea, diabetes mellitus, hypertension, and cholecystitis.

*MI occurs after
index within
follow-up*

Example 1: Subject 003 Hypertension

SDTM

CE				
USUBJID	CESEQ	CETERM	CEPRES	CEOCCUR
001	1	MYOCARDIAL INFARCTION	Y	Y
001	2	HYPERTENSION	Y	Y
002	1	MYOCARDIAL INFARCTION	Y	N
002	2	HYPERTENSION	Y	Y

*Source
RWD*

NOTES		
PT_ID	DATE	TEXT
21962	06/27/2025	PAST MEDICAL HISTORY: 1. Stage III CKD with baseline creatinine in the 2 range. 2. Status post MI on May 30, 2024 .
19251	04/16/2024	PAST MEDICAL HISTORY: Type 2 DM, hypothyroidism, hyperlipidemia, HTN , obstructive sleep apnea
19251	07/25/2024	ADMISSION DIAGNOSES: Hyperglycemia, cholelithiasis, obstructive sleep apnea, diabetes mellitus, and hypertension . DISCHARGE DIAGNOSES: Hyperglycemia, cholelithiasis, obstructive sleep apnea, diabetes mellitus, hypertension and cholecystitis.

Example 1: Subject 003 Hypertension

SDTM

CE				
USUBJID	CESEQ	CETERM	CEPRES	CEOCCUR
001	1	MYOCARDIAL INFARCTION	Y	Y
001	2	HYPERTENSION	Y	Y
002	1	MYOCARDIAL INFARCTION	Y	N
002	2	HYPERTENSION	Y	Y

*Source
RWD*

Vitals			
Patno	Date	Vital	Value
21962	06/27/2025	BP	160/115
21962	06/27/2025	BMI	33.6
19251	04/16/2024	BP	142/86
19251	04/16/2024	BMI	28.5
19251	05/28/2024	BP	140/82
19251	05/28/2024	BMI	28.2
19251	07/25/2024	BP	142/85
19251	07/25/2024	BMI	28.3

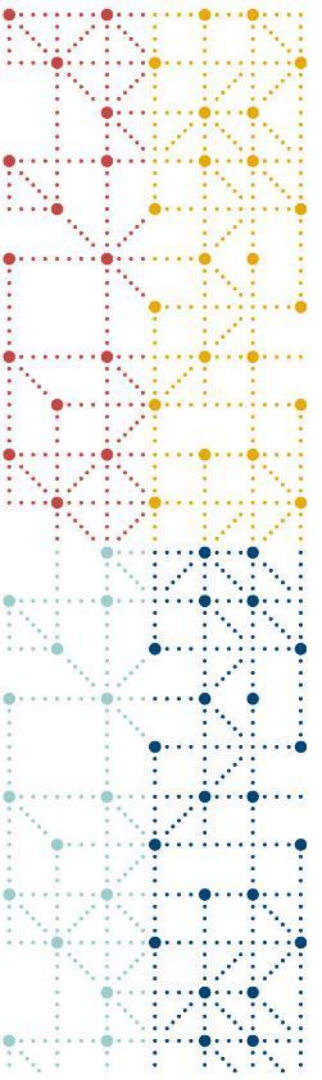
Example 1: RWD Lineage XML links & tabular view of lineage

Define-XML: <https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml>

RWD-Lineage-XML: <https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/rwd-lineage.xml>

uuid	TARGET					SOURCE							
	storage	structure	URI	rowIndex	columnName	storage	structure	URI	rowIndex	columnName	spanBegin	spanEnd	value
35060134-fc2f-4c	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	2	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	ICD10			I10
77d5b278-692e-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	2	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	TERM			Essential hyperten
7e376beb-7dad-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	2	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	DATE			06/27/2025
880af520-a197-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	3	Vital			BP
99e07ea5-bdef-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	3	Value			142/86
9a5e2706-c919-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	3	Date			04/16/2024
897e5ede-3913-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	5	Vital			BP
0a44299a-87c5-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	5	Value			140/82
b8fb983e-6d14-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	5	Date			05/28/2024
b70adc19-9e75-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	7	Vital			BP
44ead9c-8781-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	7	Value			142/85
76237584-4c5c-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	7	Date			07/25/2024
0465945a-2b61-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	2	TEXT	707	710	HTN
7c8b1ce7-dd0d-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	3	TEXT	100	112	hypertension
39c0d747-67dc-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	4	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	3	TEXT	209	221	hypertension
0f185720-823f-4e	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	1	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	1	ICD10			I21.3
de098b66-7ac9-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	1	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	1	TERM			ST elevation (ST
3fefb802-1332-4f	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	1	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	1	DATE			05/30/2025
67e89898-8383-4	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	1	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	1	TEXT	278	305	acute myocardial
8f0f595a-9d6a-4e	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	1	CEOCCUR	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/data/define/define.xml	1	TEXT	1211	1234	post MI on May 3

Tabular view: <https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example1/Example1.xlsx> (RWDLineage-Table tab)



Example 2: Elevated Liver Enzyme Adverse Event (AE)

Example 2: Elevated Liver Enzyme Adverse Event

EHR LOINC lab results → SDTM LB + AE domains

Scenario

Liver-enzyme lab panels (ALT, AST, ALP) for two subjects across two visits. Raw EHR results arrive as composite strings (e.g., "0.3507 μ kat/L"). These are parsed, converted to standard units, and mapped into SDTM LB. When results exceed normal ranges, a hepatic enzyme elevation adverse event is derived in the AE domain.

LB Transformation Pipeline

1. Direct mapping

LOINC codes → LBTESTCD/LBTEST; visit dates → LBDMTC

2. Lab value parsing

"0.3507 μ kat/L" → LBORRES (0.35) + LBORRESU (μ kat/L)

3. Unit conversion

μ kat/L → U/L stored in LBSTRES / LBSTRESU

4. Normal range evaluation

Compare to LBSTNRLO / LBSTNRHI → LBNRIND to derive AETERM
"Liver enzymes elevated above normal range"

Source Data

LOINC	Test Name	LBTESTCD
1742-6	Alanine Aminotransferase	ALT
1920-8	Aspartate Aminotransferase	AST
1775-6	Alkaline Phosphatase	ALP

Source: LabResults.csv (12 rows, 2 subjects × 3 tests × 2 visits)

AE Derivation

When a subject's ALT, AST, or ALP results exceed the normal range (LBNRIND = HIGH), a hepatic enzyme elevation adverse event is derived:

Field	Value
USUBJID	002
AETERM	Liver enzymes elevated above normal range
AESTDTC	2025-11-06

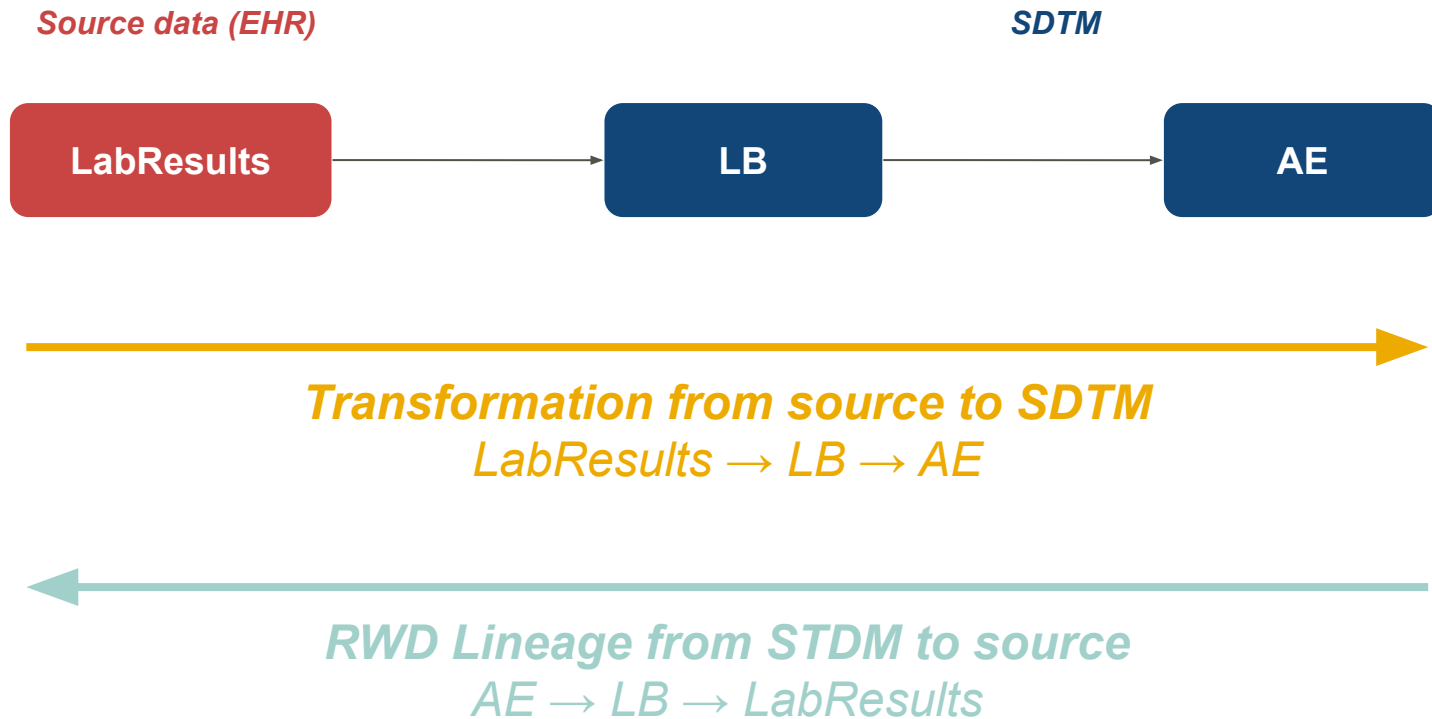
Stackable traceability: AE record traces through the LB pipeline back to the original EHR lab result strings

Example 2: SDTM AE Derivation



- **Step 1:** Map source data LabResults to SDTM LB
- **Step 2:** Unit conversion from original to standard units
 - LBORRES → LBSTRES
 - LBORRESU → LBSTRESU
- **Step 3:** Derive AE for increased hepatic enzyme
 - ALT > 55 U/L
 - AST > 48 U/L
 - ALP > 129 U/L

Example 2: SDTM AE Derivation



Example 2: Deriving AE “Hepatic enzyme increased”

SDTM

AE							
USUBJID	AESEQ	AETERM	AEDECOD	AELLTCD	AESER	AEREL	AESTDTC
002	1	Liver enzymes elevated above normal range	Hepatic enzyme increased	10060795	N	N	2025-11-06

Example 2: Deriving AE “Hepatic enzyme increased”

SDTM

AE							
USUBJID	AESEQ	AETERM	AEDECOD	AELLTCD	AESER	AEREL	AESTDTC
002	1	Liver enzymes elevated above normal range	Hepatic enzyme increased	10060795	N	N	2025-11-06

Comparison to reference range

LB											
USUBJ ID	LBSEQ	LBTEST CD	LBTEST	LBSTRES	LBSTRESU	LBSTNRLO	LBSTNRHI	LBSTNRLO	LBSTNRHI	LBSTNRLO	LBSTNRHI
002	1	ALT	Alanine Aminotransferase	2024-10-25	0.87	μkat/L	52	U/L	7	55	NORMAL
002	2	AST	Aspartate Aminotransferase	2024-10-25	0.78	μkat/L	47	U/L	8	48	NORMAL
002	3	ALP	Alkaline Phosphatase	2024-10-25	2.14	μkat/L	128	U/L	40	129	NORMAL
002	4	ALT	Alanine Aminotransferase	2025-11-06	1.00	μkat/L	60	U/L	7	55	HIGH
002	5	AST	Aspartate Aminotransferase	2025-11-06	0.82	μkat/L	49	U/L	8	48	HIGH
002	6	ALP	Alkaline Phosphatase	2025-11-06	2.19	μkat/L	131	U/L	40	129	HIGH

Example 2: Deriving AE “Hepatic enzyme increased”

SDTM

AE							
USUBJID	AESEQ	AETERM	AEDECOD	AELLTCD	AESER	AEREL	AESTDTC
002	1	Liver enzymes elevated above normal range	Hepatic enzyme increased	10060795	N	N	2025-11-06

Direct map

LB											
USUBJ ID	LBSEQ	LBTEST CD	LBTEST	LBSTRES	LBSTRESU	LBSTNRLO	LBSTNRHI	LBSTNRLO	LBSTNRHI	LBSTNRLO	LBSTNRHI
002	1	ALT	Alanine Aminotransferase	2024-10-25	0.87	μkat/L	52	U/L	7	55	NORMAL
002	2	AST	Aspartate Aminotransferase	2024-10-25	0.78	μkat/L	47	U/L	8	48	NORMAL
002	3	ALP	Alkaline Phosphatase	2024-10-25	2.14	μkat/L	128	U/L	40	129	NORMAL
002	4	ALT	Alanine Aminotransferase	2025-11-06	1.00	μkat/L	60	U/L	7	55	HIGH
002	5	AST	Aspartate Aminotransferase	2025-11-06	0.82	μkat/L	49	U/L	8	48	HIGH
002	6	ALP	Alkaline Phosphatase	2025-11-06	2.19	μkat/L	131	U/L	40	129	HIGH

Example 2: Converting LBORRES/LBORRESU > LBSTRES/LBSTRESU

LB											
USUBJ ID	LBSEQ	LBTEST CD	LBTEST	LBDTC	LBORRES	LBORRES U	LBSTRESC	LBSTRESU	LBSTNRLO	LBSTNRHI	LBNRIND
002	1	ALT	Alanine Aminotransferase	2024-10-25	0.87	μkat/L	52	U/L	7	55	NORMAL
002	2	AST	Aspartate Aminotransferase	2024-10-25	0.78	μkat/L	47	U/L	8	48	NORMAL
002	3	ALP	Alkaline Phosphatase	2024-10-25	2.14	μkat/L	128	U/L	40	129	NORMAL
002	4	ALT	Alanine Aminotransferase	2025-11-06	1.00	μkat/L	60	U/L	7	55	HIGH
002	5	AST	Aspartate Aminotransferase	2025-11-06	0.82	μkat/L	49	U/L	8	48	HIGH
002	6	ALP	Alkaline Phosphatase	2025-11-06	2.19	μkat/L	131	U/L	40	129	HIGH

Unit conversion

Example 2: Converting LBORRES/LBORRESU > LBSTRES/LBSTRESU

LB											
USUBJ ID	LBSEQ	LBTEST CD	LBTEST	LBDTC	LBORRES	LBORRES U	LBSTRESC	LBSTRESU	LBSTNRLO	LBSTNRHI	LBNRIND
002	1	ALT	Alanine Aminotransferase	2024-10-25	0.87	μkat/L	52	U/L	7	55	NORMAL
002	2	AST	Aspartate Aminotransferase	2024-10-25	0.78	μkat/L	47	U/L	8	48	NORMAL
002	3	ALP	Alkaline Phosphatase	2024-10-25	2.14	μkat/L	128	U/L	40	129	NORMAL
002	4	ALT	Alanine Aminotransferase	2025-11-06	1.00	μkat/L	60	U/L	7	55	HIGH
002	5	AST	Aspartate Aminotransferase	2025-11-06	0.82	μkat/L	49	U/L	8	48	HIGH
002	6	ALP	Alkaline Phosphatase	2025-11-06	2.19	μkat/L	131	U/L	40	129	HIGH

Unit conversion

Example 2: Mapping source data to LB

LB											
USUBJ ID	LBSEQ	LBTEST CD	LBTEST	LB DTC	LBORRES	LBORRES U	LBSTRESC	LBSTRESU	LBSTNRLO	LBSTNRHI	LBNRIND
002	1	ALT	Alanine Aminotransferase	2024-10-25	0.87	µkat/L	52	U/L	7	55	NORMAL
002	2	AST	Aspartate Aminotransferase	2024-10-25	0.78	µkat/L	47	U/L	8	48	NORMAL
002	3	ALP	Alkaline Phosphatase	2024-10-25	2.14	µkat/L	128	U/L	40	129	NORMAL
002	4	ALT	Alanine Aminotransferase	2025-11-06	1.00	µkat/L	60	U/L	7	55	HIGH
002	5	AST	Aspartate Aminotransferase	2025-11-06	0.82	µkat/L	49	U/L	8	48	HIGH
002	6	ALP	Alkaline Phosphatase	2025-11-06	2.19	µkat/L	131	U/L	40	129	HIGH

Direct map

LabResults				
PATID	LOINC Code	Lab Test	Visit Date	Lab Result
342832	1742-6	Alanine Aminotransferase	10/25/2024	0.8684 µkat/L
342832	1920-8	Aspartate Aminotransferase	10/25/2024	0.7849 µkat/L
342832	1775-6	Alkaline Phosphatase	10/25/2024	2.1376 µkat/L
342832	1742-6	Alanine Aminotransferase	11/6/2025	1.002 µkat/L
342832	1920-8	Aspartate Aminotransferase	11/6/2025	0.8183 µkat/L
342832	1775-6	Alkaline Phosphatase	11/6/2025	2.1877 µkat/L

Example 2: Mapping source data to LB

LB											
USUBJ ID	LBSEQ	LBTEST CD	LBTEST	LBDTC	LBORRES	LBORRES U	LBSTRESC	LBSTRESU	LBSTNRLO	LBSTNRHI	LBNRIND
002	1	ALT	Alanine Aminotransferase	2024-10-25	0.87	µkat/L	52	U/L	7	55	NORMAL
002	2	AST	Aspartate Aminotransferase	2024-10-25	0.78	µkat/L	47	U/L	8	48	NORMAL
002	3	ALP	Alkaline Phosphatase	2024-10-25	2.14	µkat/L	128	U/L	40	129	NORMAL
002	4	ALT	Alanine Aminotransferase	2025-11-06	1.00	µkat/L	60	U/L	7	55	HIGH
002	5	AST	Aspartate Aminotransferase	2025-11-06	0.82	µkat/L	49	U/L	8	48	HIGH
002	6	ALP	Alkaline Phosphatase	2025-11-06	2.19	µkat/L	131	U/L	40	129	HIGH

LabResults				
PATID	LOINC Code	Lab Test	Visit Date	Lab Result
342832	1742-6	Alanine Aminotransferase	10/25/2024	0.8684 µkat/L
342832	1920-8	Aspartate Aminotransferase	10/25/2024	0.7849 µkat/L
342832	1775-6	Alkaline Phosphatase	10/25/2024	2.1376 µkat/L
342832	1742-6	Alanine Aminotransferase	11/6/2025	1.002 µkat/L
342832	1920-8	Aspartate Aminotransferase	11/6/2025	0.8183 µkat/L
342832	1775-6	Alkaline Phosphatase	11/6/2025	2.1877 µkat/L

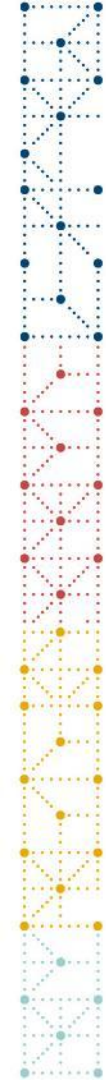
Direct map

Example 2: Mapping source data to LB

LB											
USUBJ ID	LBSEQ	LBTEST CD	LBTEST	LBDTC	LBORRES	LBORRES U	LBSTRESC	LBSTRESU	LBSTNRLO	LBSTNRHI	LBNRIND
002	1	ALT	Alanine Aminotransferase	2024-10-25	0.87	µkat/L	52	U/L	7	55	NORMAL
002	2	AST	Aspartate Aminotransferase	2024-10-25	0.78	µkat/L	47	U/L	8	48	NORMAL
002	3	ALP	Alkaline Phosphatase	2024-10-25	2.14	µkat/L	128	U/L	40	129	NORMAL
002	4	ALT	Alanine Aminotransferase	2025-11-06	1.00	µkat/L	60	U/L	7	55	HIGH
002	5	AST	Aspartate Aminotransferase	2025-11-06	0.82	µkat/L	49	U/L	8	48	HIGH
002	6	ALP	Alkaline Phosphatase	2025-11-06	2.19	µkat/L	131	U/L	40	129	HIGH

LabResults				
PATID	LOINC Code	Lab Test	Visit Date	Lab Result
342832	1742-6	Alanine Aminotransferase	10/25/2024	0.8684 µkat/L
342832	1920-8	Aspartate Aminotransferase	10/25/2024	0.7849 µkat/L
342832	1775-6	Alkaline Phosphatase	10/25/2024	2.1376 µkat/L
342832	1742-6	Alanine Aminotransferase	11/6/2025	1.002 µkat/L
342832	1920-8	Aspartate Aminotransferase	11/6/2025	0.8183 µkat/L
342832	1775-6	Alkaline Phosphatase	11/6/2025	2.1877 µkat/L

Parse lab values & units



Putting it all together...

Stack multiple stages of RWD Lineage to obtain lineage from AE to source

Example 2: Stack lineage to validate AE against source

SDTM

AE							
USUBJID	AESEQ	AETERM	AEDECOD	AELLTCD	AESER	AEREL	AESTDTC
002	1	Liver enzymes elevated above normal range	Hepatic enzyme increased	10060795	N	N	2025-11-06

LabResults				
PATID	LOINC Code	Lab Test	Visit Date	Lab Result
342832	1742-6	Alanine Aminotransferase	10/25/2024	0.8684 µkat/L
342832	1920-8	Aspartate Aminotransferase	10/25/2024	0.7849 µkat/L
342832	1775-6	Alkaline Phosphatase	10/25/2024	2.1376 µkat/L
342832	1742-6	Alanine Aminotransferase	11/6/2025	1.002 µkat/L
342832	1920-8	Aspartate Aminotransferase	11/6/2025	0.8183 µkat/L
342832	1775-6	Alkaline Phosphatase	11/6/2025	2.1877 µkat/L

Example 2: Stack lineage to validate AE against source

SDTM

AE							
USUBJID	AESEQ	AETERM	AEDECOD	AELLTCD	AESER	AEREL	AESTDTC
002	1	Liver enzymes elevated above normal range	Hepatic enzyme increased	10060795	N	N	2025-11-06

LabResults				
PATID	LOINC Code	Lab Test	Visit Date	Lab Result
342832	1742-6	Alanine Aminotransferase	10/25/2024	0.8684 µkat/L
342832	1920-8	Aspartate Aminotransferase	10/25/2024	0.7849 µkat/L
342832	1775-6	Alkaline Phosphatase	10/25/2024	2.1376 µkat/L
342832	1742-6	Alanine Aminotransferase	11/6/2025	1.002 µkat/L
342832	1920-8	Aspartate Aminotransferase	11/6/2025	0.8183 µkat/L
342832	1775-6	Alkaline Phosphatase	11/6/2025	2.1877 µkat/L

Example 2: RWD Lineage XML links & tabular view of lineage

Define-XML: <https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml>

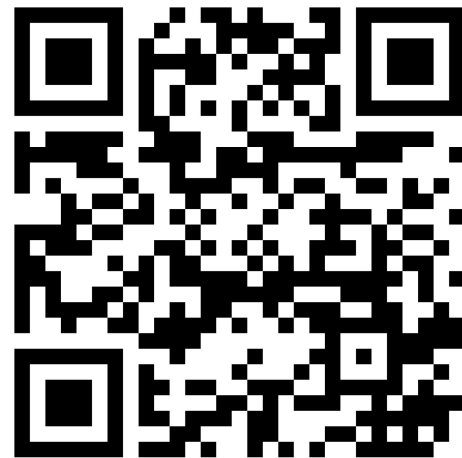
RWD-Lineage-XML: <https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/rwd-lineage.xml>

TARGET						SOURCE					
storage	structure	URI	rowIndex	columnName	value	storage	structure	URI	rowIndex	columnName	value
Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	1	AEDECOD	Hepatic enzyme increased	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	10	LOINC Code	1742-6
Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	1	AEDECOD	Hepatic enzyme increased	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	11	LOINC Code	1920-8
Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	1	AEDECOD	Hepatic enzyme increased	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	12	LOINC Code	1775-6
Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	1	AEDECOD	Hepatic enzyme increased	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	10	Lab Test	Alanine Aminotransferase
Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	1	AEDECOD	Hepatic enzyme increased	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	11	Lab Test	Aspartate Aminotransferase
Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	1	AEDECOD	Hepatic enzyme increased	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	12	Lab Test	Alkaline Phosphatase
Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	1	AEDECOD	Hepatic enzyme increased	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	10	Lab Result	1.002 µkat/L
Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	1	AEDECOD	Hepatic enzyme increased	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	11	Lab Result	0.8183 µkat/L
Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	1	AEDECOD	Hepatic enzyme increased	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	12	Lab Result	2.1877 µkat/L
Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	1	AESTDTC	2025-11-06	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	10	Visit Date	11/6/2025
Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	1	AESTDTC	2025-11-06	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	11	Visit Date	11/6/2025
Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	1	AESTDTC	2025-11-06	Filesystem	Tabular	https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/data/define/define.xml	12	Visit Date	11/6/2025

Tabular view (full lineage): <https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/Example2.xlsx> (RWDLineageFull-Table tab)

Tabular view (AE to source): <https://github.com/cdisc-org/DataExchange-RWD-Lineage/blob/main/examples/example2/Example2.xlsx> (RWDLineageAEtoSource-Table tab)

Thank You!



Join us as a volunteer!